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Notes on Statistical Mechanics' Foundations

Abstract

These notes are inspired by the presentation of Poincaré's recurrence theorem and its consequences in V. Moretti's book *Meccanica Analitica* [3]. They are divided into two parts. The first part introduces some fundamental concepts of classical (Boltzmannian) statistical mechanics (SM) and examines how certain difficulties and paradoxes in its formulation motivate the search for a more satisfactory definition of equilibrium. Its focus is therefore primarily on the underlying reasoning and conceptual structure of the theory, following the work of R. Frigg and C. Werndl, in particular their book *Foundations of Statistical Mechanics* [1]. The second part consists of an appendix containing proofs of some of the most important theorems discussed in the first part.

Dynamical systems

At the level of its micro constituents, a SM system can be characterized by the structure of a dynamical system (X, ϕ_t, μ) . X is the state space. In the case of Hamiltonian systems, X is the phase space. ϕ_t is a one-parameter group of diffeomorphisms describing the evolution and determining the trajectories in phase space. μ is the *uniform Lebesgue measure*.

Definition: Let W be the class of all physically possible worlds. The world $w \in W$ is **deterministic** if and only if $\forall w, w' \in W$, if w and w' are in the same state at some time t_0 , then they are in the same state at all times t .

This definition can be restricted to an **isolated subsystem** s of w , then considering $W_S \subseteq W$ as the set of worlds which contain counterparts of s , s is deterministic $\iff \forall w, w' \in W_S$, s and s' are in the same state at some $t_0 \implies$ they are in the same state at all times t . We require ϕ_t to be deterministic, so that every state x has exactly one past and one future; *trajectories in phase space do not intersect*.

Hamiltonian mechanics

In Hamiltonian mechanics, ϕ_t follows integral lines of solutions of the *Hamilton equations*:

$$\frac{dq_k}{dt} = \frac{\partial H}{\partial p_k}, \quad \frac{dp_k}{dt} = -\frac{\partial H}{\partial q_k}, \quad k = 1, \dots, 3m \quad (1)$$

For $m \gg 1$, it becomes too difficult to write down ϕ_t , but the form of the equations ensures that ϕ_t has certain features.

Observation: From Hamiltonian mechanics, recall

$$\frac{\partial H}{\partial t} = 0 \quad \Rightarrow \quad \left. \frac{dH}{dt} \right|_{\text{eq. of motion}} = 0 \quad \Rightarrow \quad \text{autonomous system} \quad (2)$$

Autonomous systems usually correspond to isolated systems. If the system is mechanical, H represents the **mechanical energy** E . We will restrict our attention to systems that satisfy equation (2). The constraint $H = \text{const.}$ describes a **hypersurface** X_E in X , where the time evolution is confined. Since $\phi_t : X_E \rightarrow X_E$, the hypersurface is **invariant** under ϕ_t . Moreover, regions contained in X divided by X_E are disconnected; trajectories cannot cross both of them. Consequently, each sector is mapped into itself by the dynamical one parameter group.

Theorem: Liouville's theorem. Given an autonomous hamiltonian system, if X is the phase space of the system and $A \subset X$,

$$\mu(A) = \mu(\phi_t(A)) \quad \forall A, \forall t.$$

Therefore ϕ_t is called **μ -preserving** (or measure-preserving) if it describes a solution of Hamilton's autonomous equations.

Time-reversal invariance¹

Definition: A theory T is **time-reversal invariant** if and only if every process that is allowed by the theory has its time reverse allowed too.

¹We restrict our attention to autonomous Hamiltonian systems in the rest of the notes

The definition does not imply that processes must be palindromic. A process $P(t)$ **does not need** to be mapped into itself under the time-reversal transformation $t \rightarrow -t$. For example, consider a ball moving from left to right. If we consider $t \rightarrow -t$, we are “playing” the film backwards:

$$(q, p) \rightarrow (q, -p).$$

Time reversal corresponds to replacing $t \rightarrow -t$, $p \rightarrow -p$.

In fact, Hamilton’s equations are left unchanged under $t \rightarrow -t$, $p \rightarrow -p$. The process unfolding backward in time is governed by the same laws. Therefore, what is or is not allowed by Hamilton’s equations does not change if we replace t by $-t$, so **Hamiltonian mechanics is time-reversal invariant**.

To give another example, if we evolve a point in phase space along integral curves of Hamilton equations, if $(q_i, p_i) \xrightarrow{\Delta t} (q_f, p_f)$ is allowed, then $(q_f, -p_f) \xrightarrow{\Delta t} (q_i, -p_i)$ is allowed.² Thanks to Liouville’s theorem, hamiltonian system manifest a counterintuitive behaviour. the **Poincaré recurrence**.

Theorem: Poincaré recurrence theorem. Consider a system with a state space such that $\mu(X) < +\infty$, with a dynamics ϕ_t that maps X into itself. If ϕ_t is μ -preserving, $\forall A \subset X, \mu(A) > 0$, “almost” any point will return to A infinitely many times as $t \rightarrow +\infty$.

Ergodicity

Consider a measurable function $f : X \rightarrow \mathbb{R}$, sometimes called a *phase function*. The space average is given by $\bar{f} = \int_X f d\mu$, while the time average is $f^*(x) = \lim_{\tau \rightarrow +\infty} \frac{1}{\tau} \int_{t_0}^{t_0+\tau} f(\phi_t(x)) dt$, where t_0, x are the initial conditions of the trajectory. Thanks to the **Birkhoff theorem**, $f^*(x)$ exists for almost all initial conditions and does not depend on t_0 , except for a subset of X of measure zero.

Definition: A dynamical system is said to be *ergodic* if $f^*(x) = \bar{f}$.

Since statistical mechanics deals with probabilities, it is useful to assume the measure to be *normalised*: $\mu(X) = 1$ (we can assume that without loss of generality, as for SM systems we already have $\mu(X) < +\infty$ in most cases). If we define the characteristic function given a set χ_A , we can use that as the phase function:

$$f(x) = \chi_A(x) = \begin{cases} 1 & \text{if } x \in A \subset X, \\ 0 & \text{if } x \notin A \subset X, \end{cases}$$

then

$$\bar{f} = \int_X \chi_A(x) d\mu = \int_A d\mu = \mu(A),$$

and $f^*(x)$ is the fraction of time spent in A . If the system is ergodic, $\forall A \subset X$, the fraction of time spent in A is equal to the measure of A , $\mu(A)$. As a consequence, there cannot be a set B , $\mu(B) > 0$, that the system does not visit. As a consequence, if a system has invariant hypersurfaces that divide the state space X into disconnected areas, it isn’t ergodic on X . However, the system could be ergodic on X_E or on $A \subseteq X_E$ (a hypersurface or a subset of it). If $A \subset X$ is an **invariant set** ($\phi_t(A) = A$), and the system is ergodic, $\mu(A) \in \{0, 1\}$, since the trajectory $\{\phi_t(x), t \in \mathbb{R}\} \subseteq A$, or $\{\phi_t(x), t \in \mathbb{R}\} \subseteq X \setminus A$.

Boltzmannian statistical mechanics

Definition: We define **micro-states** as points in X . **Macro-states** are then represented by subsets of X , with the characteristic that every point that belongs to them shows the same observable

²this example will be used later for the *Loschmidt objection*.

quantities. Macro-states *supervene* on micro-states: two systems cannot be in different macro-states without being in different micro-states. Therefore, we can assign uniquely a macro-state to each micro-state: $M(x)$.

Definition: Given a macro-state M , the **macro-region** of M is:

$$X_M = \{x \in X \mid M(x) = M\} \subset X.$$

Of course, we can assign a size to macro-regions using μ . If invariant surfaces X_E are present, one has to consider macro-regions that lie in X_E (and consequently consider μ restricted to X_E , giving μ_E)

Definition: The **Boltzmannian entropy** S_B of a macro-state M_α is given by:³

$$S_B(M_\alpha) = k_B \log \mu(X_{M_\alpha}) \quad \forall \alpha.$$

Coarse-graining

The idea of coarse-graining is the following. The $6m$ dimensional phase space X that represents the state of the system is "reduced" into a 6 dimensional phase space X_P , where there are m different points that bring the information of the state of the system in the initial phase space X s.⁴ The second step is to discretize the space X_P by introducing a grid, to limit the attention to the cells and their properties. We label the cells $\gamma_1, \dots, \gamma_k$.

Definition: Two points $r, s \in X_P$ are said to have the same **coarse-grained micro-state** if they are in the same cell. A specification of the coarse-grained micro-state is an **arrangement**. If we have m points, an arrangement is an ordered list that specifies which cell corresponds to each point.

We assume that all points in X_P are identical in terms of their influence on macroscopic properties, therefore the order of the list is not needed; one needs just the *distribution*.

Definition: We introduce a equivalence relation $\mathcal{R}$ between lists: given two lists r, s , $s \mathcal{R} r$ if the set of their entries is the same (up to permutations). **Distributions** are the classes of equivalence of $\mathcal{R}$.

Given $(m_1, \dots, m_k)$, the amount of points in each cell $(\delta_1, \dots, \delta_k)$, where k is the number of coarse-grained micro-states and $m = \sum_{l=1}^k m_l$ is the total number of points, the number of **arrangements** $G(D)$ of a distribution D is:

$$G(D) = \frac{m!}{\prod_{s=1}^k m_s!}.$$

Boltzmann hypothesis: Having determined the number of compatible arrangements with a distribution, the **equilibrium** corresponds to the distribution with most arrangements. Making the assumptions of non-interacting particles and energy conservation, the distribution for which the number of arrangements is maximal requires the m_i (assuming also $m_i \gg 1 \forall i$) to satisfy⁵

$$m_i = \alpha e^{-\beta E_i},$$

where E_i is the energy of a particle in cell δ_i . At this point we note that since all macro properties depend solely on D , one can say that all points in X that correspond to the same D constitute a macro-region. The "ordered" arrangements (the lists mentioned above) are not needed in the description of the macroscopic properties of a system. We now need a connection between $G(D)$ and $\mu(X_D)$, the measure of the macro-region associated with D . The connection is given by $\mu(X_D) =$

³It's defined up to an arbitrary constant

⁴namely, all the generalized coordinates and momenta of the state.

⁵it can be demonstrated imposing conservation of the number of particles and the total energy, and using the Stirling approximation

$G(D) \delta^m$, where δ is the volume of a coarse-graining cell in X_P . It can be shown that, as mentioned above $G(D)$ is maximal for the Maxwell-Boltzmann distribution (under the conditions presented). Therefore, the largest macro-region $X_D \subset X$ is the one that consists of points that correspond to that distribution. Consequently, it is justified to regard the largest macro-region as the **equilibrium macro-region**.

Objections to the combinatorial argument

Several critiques were advanced to oppose the combinatorial argument. For example, the derivation of the Maxwell Boltzmann distribution requires non-interacting particles, however such description can only be used to describe ideal gases. On a more fundamental level, coarse graining requires a specific grid, as D depends on grid cells, since in general different choices of grids lead to different results.

The latter argument brings two important questions that need to be answered separately: what justifies the choice of a grid in the first place? Why do coarse-grainings yield different results?

Starting from the first question, one justification may rely on limited knowledge, as measurement instruments have a finite resolution, so a δ can be set to match the resolution, thus providing the interpretation of coarse-graining as the reflection of our epistemic limitations.⁶

However, equilibrium properties of a system are taken to be *ontic*, and in that case they do not depend on what we know about the system (in particular, they should not depend on the coarse-graining).

One may consider it just as a mathematical ploy. The result should be preserved when taking the limit $\delta \rightarrow 0$. However, the combinatorial argument assumes $m \gg k$ and that $m_i \gg 1$ in every δ_i . But since m is fixed, if $\delta \rightarrow 0$, the strong inequalities required that are stated above will not hold when δ goes to zero. About the second question, there is no explanation why that particular coarse-graining yields the correct result.

Even beyond these two critiques, there are more problems with the use of coarse-graining. From a macroscopic point of view, we start from microstates $\{m_i\}_{i \in I}$, and postulate the existence of macrostates: $\forall x \in X, \exists! M_\alpha, \alpha \in I$ such that $M(x) = M_\alpha$, thus defining macro-regions:

$$\{X_{M_\alpha}\}_{\alpha \in I}, \quad \bigsqcup_{\alpha \in I} X_{M_\alpha} = X.$$

However, the coarse-graining argument reverses the order, constructing macro-regions without saying anything about macro-states. In particular, the connection between thermodynamic equilibrium (as the constance in time of macroscopic properties) and “combinatorial equilibrium” intended as the macro-region corresponding to the greatest number of arrangements is not trivial.

Finally, the “combinatorial equilibrium” is completely disconnected from the dynamics of the system, as it never mentions the approach to equilibrium and the time evolution of the system.

⁶One could also fix the dimension of the grid a priori, assuming it is impossible to have complete knowledge, even in principle on a scale that is smaller than δ .

The ergodic account

The gap between the fact that the Maxwell-Boltzmann distribution corresponds to the largest macro-region⁷ and the claim that the system approaches equilibrium and spends most of its time in equilibrium can be solved by assuming that the system is ergodic: the fraction of time spent in $A \subset X$ simply becomes $\frac{\mu(A)}{\mu(X)}$. For almost all initial conditions, this implies that most of the time is spent in equilibrium. However, some systems successfully described by statistical mechanics are not ergodic. Moreover, a set of measure zero apparently can avoid the equilibrium.

Loschmidt and Zermelo objections

The fact that systems can evolve out of the equilibrium macro-region is a direct consequence of the reversibility of Hamiltonian mechanics. Recalling the example given in the section about time-reversal invariance, since the transformation $(q_i, p_i) \xrightarrow{\Delta t} (q'_i, p'_i)$ is allowed, $(q_i, -p_i) \xrightarrow{\Delta t} (q'_i, -p'_i)$ is allowed too. If the macro-region that corresponds to equilibrium X_M is invariant under $q \rightarrow q, p \rightarrow -p$, systems are allowed to evolve out of equilibrium just as they can enter the macro-region. This argument is called *Loschmidt objection*. Moreover, SM systems satisfy the conditions for the Poincaré recurrence theorem to hold. As a consequence, the evolution of the system will eventually bring its state arbitrarily close to the initial state. This argument is called *Zermelo's recurrence objection*.

Residence time

The central result obtained from the combinatorial argument combined with ergodicity is that a system starting from almost every initial condition eventually spends most of its time in equilibrium. Therefore, we can re-define equilibrium to take into account these considerations. The name of the approach is **residence time**, or RT.⁸

Definition: Given the macro-state M let $LF_M(x)$ be the fraction of time that a system with initial condition x spends in the macro-region X_M in the long run (for $t \rightarrow +\infty$). Since the time average of a function g is:

$$g^*(x) = \lim_{T \rightarrow +\infty} \frac{1}{T} \int_{t_0}^{t_0+T} g(\phi_t(x)) dt,$$

we have:

$$LF_M(x) = \lim_{T \rightarrow +\infty} \frac{1}{T} \int_{t_0}^{t_0+T} \chi_M(\phi_t(x)) dt.$$

where $\chi_M(x)$ is the characteristic function of X_M . The time a system spends in X_M is the **residence time in M** .

Definition: Consider an isolated system S , described as a measure-preserving deterministic system $(X, \sigma_X, \mu_X, \phi_t)$, and let $M(x)$ be the macro-state that supervenes on micro-state $x \in X$. Let us introduce the parameter $\alpha \in (0.5, 1]$ and let $0 < \epsilon \ll 1$. If there exist a set $Y \subseteq X$ such that $\mu_Z(Y) \geq 1 - \epsilon$ and all initial states $x \in Y$ satisfy:⁹

$$LF_{\hat{M}}(x) \geq \alpha \tag{3}$$

⁷R. Frigg and C. Werndl distinguish between β -dominance and δ -prevalence as two possible meanings of "largest". In the first sense, it means that the macro-region $X_{\hat{M}}$ corresponding to equilibrium has measure $\mu(X_{\hat{M}}) \leq \beta \mu(X)$, $\beta \in (1/2, 1]$, while in the second it means that $\min_{M \neq \hat{M}} (\mu(X_{\hat{M}}) - \mu(X_M)) \geq \delta$ for a $\delta > 0$. In these notes the first sense is considered, for further reading see [2].

⁸This approach is presented in the works of R. Frigg and C. Werndl, in particular see [2]

⁹ α and ϵ can be specified to meet further requirements.

for some macro-state $\hat{M}$, then $\hat{M}$ is the α - ϵ -equilibrium state of S. Observation: the ϵ was introduced to require even weaker conditions than having a set of measure zero that does not evolve into the equilibrium.

Theorem: Dominance theorem. It can be proven that, if macro-region $\hat{M}$ is an α - ϵ -equilibrium of a system S, then:

$$\mu(X_M) \geq \alpha(1 - \epsilon) \quad (4)$$

so the equilibrium macro-region is “large”, as we expected.

Observations about RT account

There are several positive notes to be made about the RT approach. First of all, it is completely general; no assumptions about particles and their interactions are needed, and no grid in state space is needed. The new definition of equilibrium avoids problems encountered in the combinatorial argument and reflects the limitations imposed by the underlying hamiltonian dynamics. Moreover, it doesn't require ergodicity.

Finally, RT gives the general conditions for a system to have an equilibrium and provides a general existence theorem, making use of the *ergodic decomposition theorem*:

Theorem: Existence Theorem. Consider a measure-preserving system $(X, \sigma_X, \mu_X, \phi_t)$ and let $C_\omega = (X_\omega, \sigma_\omega, \mu_\omega, \phi_t)$, $\omega \in \Omega$ be its ergodic decomposition. Then there exist an α - ϵ -equilibrium iff there is a macro-state $\hat{M}$ such that for every C_ω ,

$$\mu_\omega(X_\omega \cap X_{\hat{M}}) \geq \alpha \quad (5)$$

except for components C_ω with $\omega \in \Omega'$, $\mu_X(\bigcup_{\omega \in \Omega'} X_\omega) \leq \epsilon$.

Conclusions

After briefly presenting the main concepts of Boltzmannian statistical mechanics and their roots in Hamiltonian mechanics, we discussed why some of its definitions and concepts appear insufficiently grounded, and introduced the approach developed by Frigg and Werndl, highlighting some of the issues that this reformulation addresses.

Appendix

We now intend to prove some of the key theorems presented in the first part, to show the link between the foundations of statistical mechanics and ergodic theory, a branch of mathematics devoted to study measure-preserving transformations and ergodic measures.

Theorem: Poincaré recurrence theorem¹⁰ Let $T: X \rightarrow X$ be a measure preserving transformation of a probability space $(X, \mathcal{B}, \mu)$. Let $E \in \mathcal{B}$ with $\mu(E) > 0$. Then there exists $F \subset E$ with $\mu(F) = \mu(E)$ such that for each $x \in F$ there is a sequence $n_1 < n_2 < \dots$ of natural numbers with $T^{n_i}(x) \in F$ for each i .

Proof: For $N \geq 0$ let $E_N = \bigcup_{n=N}^{\infty} T^{-n}E$. Then $\bigcap_{N=0}^{\infty} E_N$ is the set of all points of X which enter E infinitely often under positive iteration of T . If $x \in \bigcap_{N=0}^{\infty} E_N$ entered E a finite number of times, there would be a $\hat{N}$ such that $x \notin E_N \quad \forall N > \hat{N}$, but that would be absurd since x belongs to the intersection. Hence the set $F = E \cap \bigcap_{N=0}^{\infty} E_N$ consists of all points of E that enter

¹⁰the proof of this theorem and the next theorem come from [4], as their form is general. To see how the theorem can be formulated to regard hamiltonian mechanics, as well as some of its consequences, see [3].

E infinitely often. If $x \in F$, then there is a sequence $0 < n_1 < n_2 < \dots$ of natural numbers with $T^{n_i}(x) \in E, \forall i$. Then, we have $T^{n_i}(x) \in F$, since $T^{n_j - n_i}(T^{n_i}(x)) \in E, \forall j$. It remains to show that $\mu(F) = \mu(E)$. Since $T^{-1}(E)_N = E_{N+1}$ we have $\mu(E_N) = \mu(E_{N+1})$ since T is measure-preserving, and hence $\mu(E_0) = \mu(E_N), \forall N$. Since $E_0 \supset E_1 \supset E_2 \supset \dots$ we have $\mu(\bigcap_{N=0}^{\infty} E_N) = \mu(E_0)$, therefore $\mu(F) = \mu(E \cap \bigcap_{N=0}^{\infty} E_N) = \mu(E \cap E_0)$ since $E \subset E_0$.

We now introduce a theorem that is essential to prove the ergodic decomposition theorem. To do so, we must give some definitions and a theorem.

Definition: given a compact metrisable space X , we call $M(X)$ the set of probability measures over $\mathcal{B}(X)$.

Observation: It is rather easy to show that $M(X)$ is convex.

Definition: Given $T: X \rightarrow X$ a continuous transformation of X , the set of probability measures for which T is measure-preserving is denoted by $M(X, T)$.

Definition: given two measures μ_1 and μ_2 , μ_1 is absolutely continuous with respect to μ_2 , or $\mu_1 \ll \mu_2$ if for every $A \in \mathcal{B}(X), \mu_2(A) = 0 \implies \mu_1(A) = 0$

Theorem: Radon-Nikodym Theorem. Let μ, ν be two probability measures on the measurable space $(X, \mathcal{B})$. Then $\mu \ll \nu$ iff there exist $f \in L^1(\nu)$ with, $f \geq 0$ and $\int f d\nu = 1$, such that $\mu(B) = \int_B f d\nu, \forall B \in \mathcal{B}(X)$. The function f is unique almost everywhere. The function f is called the **Radon-Nikodym derivative** of μ with respect to ν and denoted by μ/ν .

Theorem: if T is a continuous transformation of the compact metric space X , then:

1. $M(X, T)$ is a compact subset of $M(X)$.
2. $M(X, T)$ is convex.
3. μ is an extreme point of $M(X, T)$ iff T is an ergodic measure-preserving transformation of $(X, \mathcal{B}(X), \mu)$

Proof

1. Suppose $\{\mu_n\}_1^{\infty}$ is a sequence of members of $M(X, T)$ and $\mu_n \rightarrow \mu \in M(X)$ (which is guaranteed since $M(X)$ is compact). Then, $\int f d\tilde{T}\mu = \int f \circ T d\mu = \lim_{n \rightarrow \infty} \int f \circ T d\mu_n = \lim_{n \rightarrow \infty} \int f d\mu_n = \int f d\mu$, so $\mu \in M(X, T)$.
2. suppose we have $\mu_1, \mu_2 \in M(X, T)$, then define $\mu = p\mu_1 + (1-p)\mu_2$: $\int f d\mu = p \int f d\mu_1 + (1-p) \int f d\mu_2$. This leads to: $\tilde{T}\mu(A) = \mu(T^{-1}(A)) = p\mu_1(T^{-1}(A)) + (1-p)\mu_2(T^{-1}(A)) = p\mu_1(A) + (1-p)\mu_2(A) = \mu(A)$, so $\mu \in M(X, T)$
3. Suppose $\mu \in M(X, T)$ and μ is not ergodic. There exist a Borel set E such that $T^{-1}E = E$ and $0 < \mu(E) < 1$. Define measures μ_1 and μ_2 by:

$$\mu_1(B) = \frac{\mu(B \cap E)}{\mu(E)}, \quad \mu_2(B) = \frac{\mu(B \cap X \setminus E)}{\mu(X \setminus E)}, \quad B \in \mathcal{B}(X).$$

Note that $\mu_1, \mu_2 \in M(X, T)$ by their definitions, and $\mu_1 \neq \mu_2$, and $\mu(B) = \mu(E)\mu_1(B) + (1 - \mu(E))\mu_2(B)$, therefore μ is not an extreme point of $M(X, T)$. Suppose $\mu \in M(X, T)$ is ergodic and $\mu = p\mu_1 + (1-p)\mu_2$ where $\mu_1, \mu_2 \in M(X, T)$ and $p \in [0, 1]$. We must show $\mu_1 = \mu_2$. In this case, we will show it by showing $\mu = \mu_1$ first. We note that $\mu_1 \ll \mu$, so that the Radon-Nikodym derivative $d\mu_1/d\mu$ exists:

$$\mu_1(E) = \int_E \frac{d\mu_1(x)}{d\mu} d\mu(x), \quad \forall E \in \mathcal{B}(X)$$

We have $d\mu_1/d\mu \geq 0$. We define $E = \left\{x \left| \frac{d\mu_1}{d\mu}(x) < 1 \right. \right\}$. We have:

$$\int_{E \cap T^{-1}E} \frac{d\mu_1}{d\mu} d\mu + \int_{E \setminus T^{-1}E} \frac{d\mu_1}{d\mu} d\mu = \mu_1(E) = \mu_1(T^{-1}E) = \int_{E \cap T^{-1}E} \frac{d\mu_1}{d\mu} d\mu + \int_{T^{-1}E \setminus E} \frac{d\mu_1}{d\mu} d\mu,$$

which gives:

$$\int_{E \setminus T^{-1}E} \frac{d\mu_1}{d\mu} d\mu = \int_{T^{-1}E \setminus E} \frac{d\mu_1}{d\mu} d\mu.$$

Since $d\mu_1/d\mu < 1$ on $E \setminus T^{-1}E$ and $d\mu_1/d\mu \geq 1$ on $T^{-1}E \setminus E$, and since $\mu(T^{-1}E \setminus E) = \mu(T^{-1}E) - \mu(T^{-1}E \cap E) = \mu(E) - \mu(T^{-1}E \cap E) = \mu(E \setminus T^{-1}E)$ (where we used the fact that T preserves μ), it must be true that $\mu(T^{-1}E \setminus E) = \mu(E \setminus T^{-1}E) = 0$. Therefore $\mu(T^{-1}E \triangle E) = 0$ so E is mapped into itself almost everywhere, and since μ is ergodic, it means that $\mu(E) = 1$ or $\mu(E) = 0$. If the first case was true, $\mu_1(E) = \mu_1(X) = \int_E \frac{d\mu_1}{d\mu} d\mu < \mu(E) = 1$ which is absurd. Hence we must have $\mu(E) = 0$. Similarly, if $F = \left\{x \left| \frac{d\mu_1}{d\mu}(x) > 1 \right. \right\}$, we get $\mu(F) = 0$ so that almost everywhere $d\mu_1/d\mu = 1 \implies \mu_1 = \mu$, which means that μ is an extreme point of $M(X, T)$.

Observation: Since $M(X, T)$ is a compact convex set we can use Choquet representation theorem to express each member of $M(X, T)$ in terms of the ergodic members of $M(X, T)$. If $E(X, T)$ denotes the set of extreme points of $M(X, T)$, then for each $\mu \in M(X, T)$ there is a unique measure τ on $\mathcal{B}(M(X, T))$ such that $\tau(E(X, T)) = 1$ and $\forall f \in C(X)$:

$$\int_X f(x) d\mu(x) = \int_{E(X, T)} \left(\int_X f(x) dm(x) \right) d\tau(m). \quad (6)$$

We write $\mu = \int_{E(X, T)} m d\tau(m)$ and call this the ergodic decomposition of μ . Hence every $\mu \in M(X, T)$ is a generalised convex combination of ergodic measures. It can be demonstrated that in our case of interest, the decomposition can be written in terms of $C_\omega = (X_\omega, \sigma_\omega, \mu_\omega, \phi_t), \omega \in \Omega$.¹¹ Then, we can prove the **Dominance theorem** and the **Existence theorem**¹².

Theorem: Dominance theorem. It can be proven that, if macro-region $\hat{M}$ is an α - ϵ -equilibrium of a system S , then:

$$\mu(X_M) \geq \alpha(1 - \epsilon) \quad (7)$$

Proof: Using the ergodic decomposition theorem, the measure-preserving deterministic system $(X, \sigma_X, \mu_X, \phi_t)$, the set X can be decomposed in the disjoint union of sets X_ω , equipped with a σ -algebra σ_ω and a probability measure μ_ω , and ϕ_t is ergodic on each $(X_\omega, \sigma_\omega, \mu_\omega, \phi_t)$. The indexing set is also a probability space $(\Omega, \sigma_\Omega, P)$, and for any square integrable function it holds that:

$$\int_X f d\mu_x = \int_\Omega \int_{X_\omega} f d\mu_\omega dP. \quad (8)$$

Suppose that the system has an α - ϵ -equilibrium $\hat{M}$. Using the ergodic decomposition on $f(x) = \chi_M(x)$, where M is an arbitrary macro-state, yields:

$$\mu_X(X_M) = \int_X \chi_M(x) d\mu_x = \int_\Omega \int_{X_\omega} \chi_M(x) d\mu_\omega dP. \quad (9)$$

For an ergodic system $(X_\omega, \sigma_\omega, \mu_\omega, \phi_t)$, the time average is:

$$LF_M(x) = \int_{X_\omega} \chi_M(x) d\mu_\omega = \mu_\omega(X_M \cap X_\omega).$$

¹¹For a proof, see [5] (since the space X is a compact metric space, it is also complete and separable, therefore the proof covers our case).

¹²These proofs can be found in [2].

From the definition of α - ϵ -equilibrium and because ϕ_t acts ergodically on each $(X_\omega, \sigma_\omega, \mu_\omega)$, for almost all $x \in X_\omega$, $X_\omega \subseteq Y$:

$$\alpha \leq LF_{\hat{M}}(x) = \int_{X_\omega} \chi_{\hat{M}}(x) d\mu_\omega. \quad (10)$$

We can assume that $X_\omega \subseteq Y$ because since the motion restricted to a X_ω is ergodic, either $\forall x \in X_\omega, LF_{\hat{M}}(x) \geq \alpha$ which means that $X_\omega \subseteq Y$, or the fraction of time spent is not greater than α , then $X_\omega \subseteq X \setminus Y$. Then we are just interested in the X_ω for which $X_\omega \subseteq Y$. Then since $\mu_X(Y) = 1 - \epsilon$:

$$\alpha(1 - \epsilon) = \alpha\mu_X(Y) = \int_{\Omega'} \alpha dP = \int_{\Omega'} \mu_\omega(X_{\hat{M} \cap X_\omega}) dP \leq \mu_X(X_{\hat{M}}) \quad (11)$$

Theorem: Existence Theorem. Consider a measure-preserving system $(X, \sigma_X, \mu_X, \phi_t)$ and let $C_\omega = (X_\omega, \sigma_\omega, \mu_\omega, \phi_t)$, $\omega \in \Omega$ be its ergodic decomposition. Then there exist an α - ϵ -equilibrium iff there is a macro-state $\hat{M}$ such that for every C_ω ,

$$\mu_\omega(X_\omega \cap X_{\hat{M}}) \geq \alpha \quad (12)$$

except for components C_ω with $\omega \in \Omega'$, $\mu_X(\bigcup_{\omega \in \Omega'} X_\omega) \leq \epsilon$.

Proof:

$\boxed{\Rightarrow}$: Assume that there exist an α - ϵ -equilibrium. We can use again the ergodic decomposition into $C_\omega = (X_\omega, \sigma_\omega, \mu_\omega, \phi_t)$. The definition of α - ϵ -equilibrium implies that an equilibrium macro-state exist a $\hat{M}$ such that: for almost all $x \in X_\omega$ and for all components C_ω , except at most components $C_\omega, \omega \in \Omega'$ with $\mu_X(\bigcup_{\omega \in \Omega'} X_\omega) \leq \epsilon$:

$$\alpha \leq LF_{\hat{M}}(x) = \int_{X_\omega} \chi_{\hat{M}}(x) d\mu_\omega = \mu_\omega(X_{\hat{M} \cap X_\omega}) \quad (13)$$

which is the desired claim.

$\boxed{\Leftarrow}$: Suppose that there exist a macro-state such that, given the ergodic decomposition components C_ω , it holds that

$$\mu_\omega(X_\omega \cap X_{\hat{M}}) \geq \alpha,$$

except for components C_ω with $\omega \in \Omega'$ such that $\mu_X(\bigcup_{\omega \in \Omega'} X_\omega) \leq \epsilon$. Then, by defining Y :

$$Y = X \setminus (\bigcup_{\omega \in \Omega'} X_\omega) \implies \mu_X(Y) = 1 - \epsilon.$$

Then we have:

$$\alpha \leq \mu_\omega(X_{\hat{M}} \cap X_\omega) = \int_{X_\omega} \chi_{\hat{M}}(x) d\mu_\omega = LF_{\hat{M}}(x),$$

Which means that $\hat{M}$ fulfills the definition of a α - ϵ -equilibrium.

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